Band-1 Computational Experiment: Hilbert 16th problem — quadratic vector fields (literature effective bounds + interval/$ heta$-tube Poincaré probe)

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Band-1 Computational Experiment: Hilbert 16th problem — quadratic vector fields

Honesty stamp. This note does **not** claim Hilbert's 16th problem solved. It does **not** claim a uniform finite bound $H(2) < \infty$. It does **not** carry a Clay or finished-ToE stamp. It records the standing literature ledger for quadratic planar fields, evaluates one published **restricted** effective bound at concrete (δ, σ, κ), and runs a **reproducible enclosure experiment** on **two** named normalized quadratics (center-distance via `mpmath.iv`; Poincaré displacement via RK4 step-doubling + θ-dependent tube Lipschitz + Gronwall).

GATE-04: LOCKED · no Clay · no ToE.

Abstract

Hilbert's 16th problem asks for the maximum number $H(n)$ of limit cycles of a planar polynomial vector field of degree n. For $n = 2$ the standing facts are: $H(2) \geq 4$ (Shi; Chen–Wang); each individual quadratic has finitely many limit cycles (Bamón); local cyclicity of a quadratic focus/center is at most 3 (Bautin); a **uniform** $H(2) < \infty$ remains open (Dumortier–Roussarie–Rousseau program). Ilyashenko–Llibre give an explicit but astronomical bound on δ-tame cycles for fields σ-distant from centers and κ-distant from singular quadratics.

Tonight's Band-1 probe:

1. locks the literature ledger (proved vs open vs conjecture; 2024–2026 survey notes);
2. evaluates $H(2, \delta, \sigma, \kappa)$ at $(\delta, \sigma, \kappa) = (0.05, 10^{-3}, 0.05)$;
3. interval-encloses Zoladek/Ilyashenko–Llibre center quantities $g_1, \ldots, g_4$ for named normalized quadratics **Q1** and **Q2**;
4. certifies, as **validated numerics**, one sign-change bracket of the Poincaré displacement on a radial section for **each** of **Q1** and **Q2** (existence of ≥ 1 surrounding limit cycle for that field only).

Problem statement

Quadratic Hilbert number. Let $H(2)$ be the maximal number of isolated periodic orbits (limit cycles) of a real planar quadratic vector field

$$\dot{x} = P_2(x, y), \qquad \dot{y} = Q_2(x, y).$$

- **Open target (Band 1):** obtain any uniform effective upper bound on $H(2)$, or complete the DRR finite-cyclicity program.
- **Finite checks (this note):** literature ledger + restricted-bound evaluation + two named-field enclosure experiments.

Literature ledger (what is proved vs open)

Fact	Status	Source
$H(2) \geq 4$	**LITERATURE / examples**	Shi, *Scientia Sinica* **23** (1980); Chen–Wang (1979)
Each quadratic has finitely many LCs	**THEOREM** (not uniform)	Bamón, *Publ. IHES* **64** (1986)
Focus/center cyclicity ≤ 3 in quadratic family	**THEOREM**	Bautin (1952/1954)
Uniform $H(2) < \infty$	**OPEN**	DRR program: finite cyclicity of 121 graphics (Dumortier–Roussarie–Rousseau, *JDE* **110**, 1994)
$H(2) = 4$	**CONJECTURE** (not proved)	Widely stated; **not** adopted here
Restricted effective bound $H(2, \delta, \sigma, \kappa)$	**THEOREM** (restricted class)	Ilyashenko–Llibre, arXiv:0910.3443
Variational arXiv claim $H(2) = 4$	**NOT ELEVATED**	arXiv:2103.07193 — not peer consensus
General Dulac/Ilyashenko finiteness	**UNDER DISCUSSION**	Recent surveys note possible gaps; Bamón remains the quadratic citation

Standing summary adopted here (2026-09):

$$\boxed{4 \leq H(2) \leq ? \qquad \text{(uniform upper bound unknown)}}.$$

Finiteness known; effective bounds live. Per-field finiteness for quadratics is classical (Bamón). What remains open is any *uniform* effective upper bound on $H(2)$ across all coefficient space. The only published estimate of an "effective" style is the Ilyashenko–Llibre *restricted* bound below — finite but astronomical, and not uniform.

Restricted Ilyashenko–Llibre bound (literature; evaluated tonight)

For normalized quadratics that are σ-distant from centers and κ-distant from singular quadratics, the number of δ-tame limit cycles satisfies

$$H(2,\delta,\sigma,\kappa) = |\log\sigma|\ \exp\bigl(\exp(10^{25}\,\delta^{-31}\kappa^{-2})\bigr)$$

for $(\delta,\sigma,\kappa) \subset (0,0.1)^3$ (Ilyashenko–Llibre, Theorem 5). The authors call the estimate "irrealistic but the only known estimate of this kind." **It is not a uniform bound on $H(2)$.**

Tonight's concrete evaluation:

$$(\delta,\sigma,\kappa) = (0.05,\ 10^{-3},\ 0.05) \quad\Rightarrow\quad \text{inner exponent } 10^{25}\delta^{-31}\kappa^{-2} \approx 8.59\times 10^{67},$$

hence $H(2,\delta,\sigma,\kappa)$ is **finite but astronomical** (tower beyond practical enumeration). Claim level: **LITERATURE formula + float arithmetic on parameters**.

Named systems Q1 and Q2

Normalized complex form (Ilyashenko–Llibre cell $A = 1$):

$$\dot z = \mu z + z^2 + Bz\bar z + C\bar z^2, \qquad \mu = \lambda_1 + i.$$

Field	λ_1	B	C
Q1	0.05	$0.4 + 0.35\,i$	$0.15 - 0.25\,i$
Q2	0.03	$0.50 + 0.20\,i$	$0.10 - 0.30\,i$

Normalization: both satisfy $|B| \le 2$, $|C| \le 1$, $|A| = 1$.

Polar reduction ($f = \operatorname{Re} h$, $g = \operatorname{Im} h$, $h = Ae^{i\theta} + Be^{-i\theta} + Ce^{-3i\theta}$):

$$\frac{dr}{d\theta} = r\frac{\lambda_1 + rf(\theta)}{1 + rg(\theta)}.$$

Limit cycles surrounding the origin correspond to isolated positive zeros of the displacement $P(r) - r$ of the 2π-Poincaré return on a radial section.

Method

A. Interval center-distance (`mpmath.iv`)

Center quantities (Zoladek / Ilyashenko–Llibre):

$$\begin{aligned} g_1 &= \lambda_1, \\ g_2 &= \operatorname{Im}(AB), \\ g_3 &= \operatorname{Im}\bigl[(2A + \bar B)(A - 2\bar B)\bar B\, C\bigr], \\ g_4 &= \operatorname{Im}\bigl[(2A + \bar B)(|B|^2 - |C|^2)\bar B^2 C\bigr]. \end{aligned}$$

Each real coefficient is widened by $\pm 10^{-15}$; $\sum |g_j|$ is enclosed. Claim level: **INTERVAL ARITHMETIC**.

B. Distance to singular quadratics

With $b = B - \mu/\bar{\mu}$, $c = C$, set $\kappa_{\text{dist}} = \sqrt{|b|^2 + |c|^2}$ and take $\kappa = 0.05 < \kappa_{\text{dist}}$. Claim level: **FLOAT ARITHMETIC on named coeffs**.

C. θ-tube Poincaré enclosure (validated numerics)

For each section radius r_0:

1. integrate $dr/d\theta$ by RK4 with N and $N/2$ steps (step-doubling residual);
2. build a θ**-dependent** tube $|r - r_{\text{num}}(\theta)| \leq$ pad (not a crude rectangle in r);
3. bound $L = \max|\partial F/\partial r|$ on the tube with $|1 + rg| \geq 0.05$;
4. enclose $P(r_0) - r_0$ by residual $\times e^{2\pi L}$;
5. declare a **sign-definite** enclosure if the interval excludes 0.

Opposite signs on adjacent r_0 yield a continuous-root bracket (existence of ≥ 1 LC for that named field). Claim level: **VALIDATED NUMERICS** (not a Lean/Isabelle certificate; float RK4 with a posteriori Gronwall).

D. Bautin / weak-focus sister

For the $\lambda_1 = 0$ sister of each named field, enclose $g_2 = \text{Im}(AB)$ and the jet proxy $a_3 = (-2\pi)g_2$. Nonzero g_2 excludes a center; Bautin's theorem then limits small Hopf cyclicity by 3. Claim level: **INTERVAL + LITERATURE**.

Method limits

- No claim that $H(2) < \infty$.
- No new world-record lower bound beyond Shi / Chen–Wang ($H(2) \geq 4$).
- Restricted IL bound is not uniform.
- θ-tube enclosures are for two named fields; they do not bound all quadratics.
- Step-doubling residual is a standard a posteriori estimate, not a machine-checked interval ODE integrator (CAPD/Ariadne-class).

Experiment design

Parameter	Choice
Probe	Hilbert 16 / quadratic effective bound + enclosure
Slug	`hilbert16_quadratic` (extends `hilbert16_quadratic_interval`)
Host	Linux box, Python 3.13 + NumPy + mpmath (venv `.venv_h16`)
Named fields	`Q1`, `Q2` (IL-normalized)
Success criterion	Ledger locked; IL bound evaluated; σ-distance PASS on both; ≥ 1 sign bracket PASS per field; no overclaim

Parameter	Choice
Wall	≈ 66 s

Code: `/workspace/experiments/band1/hilbert16_quadratic/verify_h16_quadratic.py`
Lab mirror: `/workspace/mission-control/labs/hilbert16_quadratic_interval/`
Raw JSON: `results.json`

Results (numbers!)

A. Literature / restricted bound

- Standing: $4 \le H(2)$ with uniform upper **open**.
- IL evaluation at $(\delta, \sigma, \kappa) = (0.05, 10^{-3}, 0.05)$: inner $10^{25}\delta^{-31}\kappa^{-2} \approx 8.59 \times 10^{67}$ $\to$ **FINITE_BUT_ASTRONOMICAL**.

B. Center quantities (`mpmath.iv`)

Field	$\sum \|g_j\|$ enclosure	σ-distant	κ_{dist}
Q1	$[0.574739328124991, 0.574739328125009]$	**PASS** ($\ge 10^{-3}$)	≈ 1.447
Q2	$[0.291542999999993, 0.291542999999997]$	**PASS** ($\ge 10^{-3}$)	≈ 1.538

Both: all of $g_1, \dots, g_4$ exclude 0 under $\pm 10^{-15}$ widening; κ-distant PASS vs $\kappa = 0.05$.

C. Poincaré displacement (θ-tube)

Q1 selected enclosures (displacement $P(r) - r$):

r_0	Displacement interval	Sign
0.365	$[+6.110 \times 10^{-4}, +6.125 \times 10^{-4}]$	$+$
0.37	$[-1.794 \times 10^{-3}, -1.792 \times 10^{-3}]$	$-$

Proved sign bracket (validated numerics): $r \in [0.365, 0.37]$ with $d(0.365) > 0 > d(0.37)$.

Q2 selected enclosures:

r_0	Displacement interval	Sign
0.38	$[+2.968 \times 10^{-3}, +2.970 \times 10^{-3}]$	$+$
0.40	$[-3.578 \times 10^{-3}, -3.574 \times 10^{-3}]$	$-$

Proved sign bracket (validated numerics): $r \in [0.38, 0.40]$ with $d(0.38) > 0 > d(0.40)$.

Hence **each** of **Q1** and **Q2** has **at least one** limit cycle surrounding the origin intersecting its certified radial interval.
Not uniqueness proofs; **not** a contribution to uniform $H(2)$.

D. Weak-focus sisters

- **Q1** sister: $g_2 \in [0.35 \pm 10^{-18}]$ excludes 0.
- **Q2** sister: $g_2 = \mathrm{Im}(AB) = 0.20$ excludes 0 under interval widening.
- Combined with Bautin: literature cyclicity ≤ 3 at a quadratic focus; tonight only certifies "not a center" for each sister.

What is proved vs numerical vs literature

Claim	Level
$H(2) \geq 4$	Literature examples
Uniform $H(2) < \infty$	**Open** (not proved here)
Bautin cyclicity ≤ 3	Literature theorem
IL formula for $H(2, \delta, \sigma, \kappa)$	Literature theorem (restricted)
Numeric value of IL tower at tonight's (δ, σ, κ)	Float evaluation of published formula
Q1, **Q2** are σ- and κ-distant	Interval / float on named coeffs
Q1 has ≥ 1 surrounding LC in $[0.365, 0.37]$	Validated numerics (θ-tube)
Q2 has ≥ 1 surrounding LC in $[0.38, 0.40]$	Validated numerics (θ-tube)
Hilbert 16 solved / Clay / ToE	**FORBIDDEN — not claimed**

Reproducibility

```
python3 -m venv .venv_h16 && .venv_h16/bin/pip install mpmath numpy
.venv_h16/bin/python experiments/band1/hilbert16_quadratic/verify_h16_quadratic.py
```

Expected one-liner matches `results.json` → `summary_one_liner`. Wall ≈ 66 s on the box.

Conclusion

Band-1 Hilbert-16 / quadratic lane delivers a **concrete finite computation** without overclaim: the only known style of **effective** bound (Ilyashenko–Llibre restricted) is evaluated and confirmed astronomical; independent **interval + θ-tube** experiments on named systems **Q1** and **Q2** each prove a displacement sign bracket and hence ≥ 1 limit cycle for that field. Finiteness per quadratic is known (Bamón); **effective uniform** $H(2) < \infty$ remains open / live.

ZoraASI / C.M. Baird — Band-1 Hilbert 16 quadratic — 2026-09-11.